Code No: 134AP

R16

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year II Semester Examinations, March - 2022
DATABASE MANAGEMENT SYSTEMS
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Differentiate between relation schema and relation instance.
b) Define the terms Arity and Cardinality of a Relation.
c) What are Domain Constraints? Explain. [5+5+5]

2.a) Discuss about E-R model and its additional features.
b) What are the function of DBA? Explain about different types of database users. [8+7]

3.a) Illustrate about Outer Joins with examples.
b) Explain about Nested queries and correlated nested queries. [7+8]

4.a) Explain the differences between Tuple Relational Calculus and Domain Relational Calculus.
b) Discuss about Triggers and Active Databases. [9+6]

5.a) Define Functional dependencies and explain how are primary keys related to FD's.
b) Illustrate about inclusion dependencies with examples. [8+7]

6.a) Describe the properties of Decomposition. Demonstrate with examples.
b) Explain about join dependencies and Fifth Normal Form. [8+7]

7.a) What are the issues in implementing atomicity and durability? Discuss.
b) Illustrate concurrent execution of transactions with examples. [7+8]

8.a) What are the differences between hash based indexing and Tree based indexing?
b) Explain about the organization of records in files. [9+6]

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